

Semiconductor & AI Industry – Credo Technology & Astera Labs – Design Verification Leader at Microsoft Corp

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Specialist Background

- ▶ Over 20 years' of experience in the semiconductor industry, with strong expertise in the AI infrastructure space
 - ▶ Well-placed to discuss the semiconductor industry, broader AI ecosystem, requirements in data centre and the interconnect market, as well as a semiconductor company that could benefit a lot from AI
- Strong familiarity with Credo Technology and Astera Labs' products and future competition, as well as which one is becoming a larger part of deployments and has more competition coming in

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Transcription begins

Analyst:

Are you familiar with the requirements in data centres in terms of interconnects, optical and whatnot, or are you more about the chips? Where do you spend most of your time?

Specialist:

Yes. I have years of experience. I worked almost all layers (ph). Initial of my career is basically more hands-on the chip design architecture development phase and then now basically on the tuning of the workload, what kind of workload will fit into particular chips, particular data centres because now you have like multiple cases. Before we used to build the data centre using the CPUs. Now we have CPU, GPU, XPU, TPU [sic], all those infrastructure and basically trying to figure it out like what kind of role or workload will be handled by what kind of compute so that it's more efficient and more cost-optimised solutions.

Analyst:

Let's talk on the chip side. The recent news is that Google has been building the TPU [tensor processing unit] for several years. We still think that TPU is probably two generations behind the Nvidia ones. I know that Amazon is building on the Trainium side, and then there are a few other chip companies, such as Groq, that are coming up. Is Microsoft building its own chip, or is there a popular name for it? I've not heard about it. If not, where are they thinking about it from a chip standpoint?

Specialist:

I would say like every software company, where if they want to be competitive in the market, there is no option. You have to build your own silicon. That's where you see every single company is basically developing the silicon. You simply cannot keep using the GPU where the original intent of the GPU is not even meant for training or inference workload. Since there was no solution, people started using it. Then over a period of time, you have to find some solution, where you consume the less energy, you develop a cheaper alternative. At the same time, you basically get the same performance. Definitely, there is a huge demand for this custom ASIC, and that's where you see the industry is already moving, swiftly from GPU workload to the customised GPU from various angles.

Analyst:

On that front, do you feel that Trainium is succeeding? Do you think that Trainium is not able to get there? Do you have any thoughts on Meta? Is Meta going to use Google's TPUs? Is it building on? What are you hearing in the industry?

Specialist:

Yes. Here is the thing. Google is the first one who envision that we cannot rely on off-the-shelf compute because they were the search engine and they came to know that just to run the one query, it basically consumes that much compute and that much energy. That's why they started building the TPU. They

started building TPU almost from 2014-15 timeframe. It's more than a decade now. Now you see the news is coming to the market. If you look at the team, it's not like they overnight built the team. They are working on last 10 years and they reach to the version 6 TPU. That's why the use case that basically the initial plan was Google to use for the internal workload, which is very well aligned so that whatever workload, they optimise their chip, they use it so that they don't have to pay to the Nvidia. Then what happened is the companies like Meta, Amazon, all those companies, they also acquired and they are also trying to follow it. What happened is they are late compared to the Google because they started after that. Also the focus was not that much like you see the way the Google was continuously innovating and improving. Over there, it was a little slow.

Now the Google has already the GPU competitive alternative where you can see the performance at less price. Now Meta knew, I don't know whether it's 100% confirmed or not, but it does make sense. For example, today, the Meta may be using some, let's say, 10% of their own chip, then they will be buying 90% from outside. Outside means mostly Nvidia and Intel, AMD, and all. Now they are getting another alternative from Google, which they are saying they are claiming that's much cheaper alternative. If it is solving the need for Meta while they improve their own silicon infrastructure, I think that is the reason that they are moving, diversifying from Nvidia to Google TPU now. Whether they will build their own or not, there is no doubt about it. They have silicon team, and they are going aggressive. They are building it, and they have even outside data centre also like silicon for various new ICs. Now when you talk to Amazon, Amazon also the same thing. They acquired (inaudible) and they started building. They have silicon in the new IC side, they have silicon from the data centre infrastructure side. They are also trying to deploy and trying to sell, and they are also claiming themselves as an alternative to Google TPU as well as Nvidia GPU. Everybody is in this game. Now what time it will be really deployed at what volume. That's the only question.

Analyst:

Are you familiar with the interconnect market? Do you know much about companies such as Credo Technology or Astera Labs?

Specialist:

Yes.

Analyst:

What's your high-level view on Credo Technology and Astera Labs in terms of products and competition in the future? Can you share your views on the next 12 months or so, like how things will change?

Specialist:

Yes. See, here is the thing that these are the companies, the Credo and Astera Labs both. If you see the current Intel CEO, Lip-Bu Tan was the Board of Director in both of these. If you look at his history, his vision, he basically have established 100-plus start-up in the silicon. He has invested, he has bring up those start-ups himself to the success level. Now when this time came for the AI infrastructure, where every company is basically trying to innovate and solve the bottleneck into the data centre interconnect, these two companies came into the market. Now when you look at the Credo, so the Credo, mainly they are focusing on the high-speed short-reach and DSP interconnect technology. They are basically for the AI network and hyperscaler infrastructure. They have a very good portfolio in terms of short-reach IP, short-reach chiplets, some sort of pies and things like that. They have multiple products in this side.

Again, the company is very small, but it is basically going aggressive and multiple big customers are basically using their solutions into the data centre. That is the reason that you see the huge success into the Credo, and it doesn't look like it will slow down anymore because you hear every single day more and

more investment going into the data centre. This company is definitely a good leader and coming up with all sort of solutions that goes into the data centre connectivity.

Analyst:

Are you talking to customers directly to see whether, let's say, before Credo, what they would use? What percentage of their infrastructure today is going to the active electrical vs normal cables?

Specialist:

If you look at before the AI kind of data centre boom, there was not this level of bigger compute interconnect required. The speed was lower. The regular interconnect like Ethernet base or any other interconnect would be good enough in the data centre because if you look at the compute that goes inside the rack and number of racks itself were much, much lower in the non-AI traditional workload. Now this thing came into the picture that the industry have innovated the compute speed, they have innovated the memory and I/O speed. Now everything is moving almost like 4-5x faster than what used to happen four years back. With that, you can't have an interconnect where you are always bottlenecked through the bandwidth and latency transfer. This became a really, really strong point for the Credo and they're basically aiming up with this infrastructure, the solution, and they are leader in terms of providing early to the market and competitively to the market. They are also in various kind of cables also because a lot of cables go into the data centre and they do that development as well.

Analyst:

Before the recent AI boom, you didn't necessarily need the power that Credo offers.

Specialist:

Yes, definitely. There was another alternatives like you can use the switches from Cisco, you can use some other switches from Broadcom, and you can have some standard connectivities for the data centre kind of thing, but not at the level that you see it today, based on the high end where you have like thousands of GPUs talking each other in one rack and one pod and things like that. The scale is basically increasing super-fast. It's basically about the scale. Like before, if you look at the rack, in one rack, whatever the compute capacity you have four years back, it is now 1,000x more because chips itself is becoming more and more complex and chips have power is more and more complex, capacity is more and more complex. Then on the other side, you are moving faster into the PCI bandwidth, memory bandwidth, the speed. All those things are pretty super-fast. You move from one generation to another generation, you just get double. Then you increase your number of cores. For example, before A6032 cores are really good scalable architecture. Now you see people have like 64 cores, 1,024 cores. It's going above and beyond. Then you are basically connecting cluster to cluster, socket to socket. It's basically both way, scale up and scale, horizontal and vertical, both expansion is happening.

Analyst:

Do you think Credo is gaining share within hyperscalers, or is the percentage of deployments tied to Credo remaining steady, but it's still growing?

Specialist:

Yes, definitely. Hyperscalers are using the solutions from the Credo, and it is growing.

Analyst:

Do you think, as a percentage of the pie, Credo is growing within that, or is it growing with the pie itself becoming larger?

Specialist:

Both, I would say, both. The one thing is basically the demand and another thing is how much they can

supply from the capacity point of view. Demand is there, and they are basically increasing their capacity at a faster pace. It's growing at both the level.

Analyst:

What about Astera Lab in terms of product and future competition?

Specialist:

Yes. For the Astera Lab, if you look at it, they basically started with the tiny product called retimer. Basically, they envision that, when there is a data centre and when the speed goes increase, then basically the quality of the signal basically degrade over a period of time in a long distance (audio distorts). That's where you basically need a retimer. The retimer is a very tiny component that goes into the data centre of the PCI Express-related solutions. They develop that. Basically, the companies started using it. That was one product, and then they started developing like CXL-related solutions, PCIe-related solutions, some cables, retimers and then they are going behind even optics. They are basically enabling the full PCIe connectivity across the rack. Basically, they have a solution to basically connect anything to anything like CPU to CPU, CPU to GPU, GPU to GPU, all kind of connectivity solutions they have implemented and already deployed to the real data centre now. They have their own market, and they are playing very big into that and very successful team basically working on this.

Analyst:

If you had to compare the last vs the next 12 months between these two companies, which one do you think is becoming a larger part of deployments?

Specialist:

Both of them are there. Their product and their focus is not exactly same. They are competing in some of the areas, but then it's not exactly the same. Both of these are leading player and very successful so far. Also, they are going QoQ delivering better and superior solutions. There is room for both of them to grow.

Analyst:

Which one do you think has more competition coming in?

Specialist:

Competition-wise, I would say the Astera Labs would have more competition compared to Credo because they are competing directly with the Broadcom. They are competing with bigger players. That doesn't mean that they will not be able to compete with them. Competition-wise, I feel Astera Labs has a more competition.

Analyst:

Between Marvell Technology and Broadcom, we all know Broadcom is bigger and going to have more design wins. Do you feel Marvell is losing or can it win as well in the next 12-18 months?

Specialist:

See, here, we are talking about the data centre semiconductor business. When everything grows above and beyond the capacity, then in that kind of market, like everybody has a room to play because, for example, they may not be able to take the high-end business, but still they will be able to take a second-level business, mid-tier business, low-tier business. In that way, there is definitely a good potential for Marvell to play in this business, in this role. Your question is next 12-18 months. I don't see anything like a slowdown in this semiconductor space, at least for next two years. Definitely, they have a room to play from these matters. Now let's say, if things changes, and basically, let's say, the Broadcom is able to take each and every single order that is available from the market. At the time, they have more chances to

basically win. That means whoever is number two, the chances will basically reduce to win the business. They may even lose the business. Right now, I think not even number one, number two, number three player also has a lot of room to grow and provide the solutions to the industry.

Analyst:

Let's say, the overall demand from a CAPEX standpoint, from a data centre standpoint, we know that things are growing pretty fast. Let's say, right now, we may assume that we're growing 40% YoY. Is your sense that next year, the 40% becomes 50%, or 40% becomes 30%? It's still growing, but do you feel the growth slows down or grows a little? How do you think from that percentage standpoint?

Specialist:

Growth up.

Analyst:

You're thinking that 40% at least goes to 45%?

Specialist:

Yes, percentage of growth is still growing.

Analyst:

Are there companies that you deal with that you feel like should benefit the most, or maybe they're not as popular today, but maybe in 12 months, there are segments that are going to become more important?

Specialist:

From the company's point of view?

Analyst:

Yes, company, segment or maybe semiconductor, whatever is your purview, what do you think will become even more important 6-12 months from now?

Specialist:

One thing is, from my view is, in the AI infrastructure, there are multiple layers, like starting from the energy infrastructure, chips, basically then models, all the applications and all. Right now, I think all the hyperscalers are basically still in a development phase where they are increasing their CAPEX to basically build the capacity. One can continue doing that if they see that level of demand. From that point of view, I think the sector-wise is basically the companies related to chips, like Broadcom, Nvidia, AMD, Marvell. These kind of companies will basically continue winning. Smaller companies like Astera Labs, Credo and another one is Arteris basically develop the NoC. That is also growing pretty fast. These companies will basically be growing. Then basically, companies like those who have a foundry business like TSMC, GlobalFoundries, those business will be growing as well. Then companies who basically supply the equipment into those foundries like ASML, those will be growing too.

Then another area is basically the energy side, the companies who basically supply the energy to the data centres or basically the equipment to the data centre. Those will be growing like Schneider Electric, the companies who are the manufacturer of the UPS, inverters, those kind of things. Then it will be basically moved, let's say, if everything goes well in the next 18 months to two years, then basically, it will be moved to the companies who basically are developing the solutions, like all the hyperscalers and big companies because they are the one who spend so much building those. Eventually, their solutions will reach to the customers.

Analyst:

Do you know a company called SiTime? It's a semiconductor company timing chips.

Specialist:

No, I'm not familiar.

Analyst:

Is there any other semiconductor company outside of the chips that you feel should benefit a lot from AI?

Specialist:

There is a company called Navitas Semiconductor. That is also a very good company can get a potential from this AI wave.

Analyst:

What does it do that's distinctive?

Specialist:

They are basically in developing the products, and that goes to the basically the data centre-related thing. The NVTs, they have some gallium nitride related the product that goes into the manufacturing.

Analyst:

Where have you seen the company being used?

Specialist:

The NVTs, I haven't worked myself with them based on their solutions and all. It's basically used in the manufacturing side.

Transcription ends

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